

ARTIFICIAL INTELLIGENCE

Lecture Set 16
Decision Trees

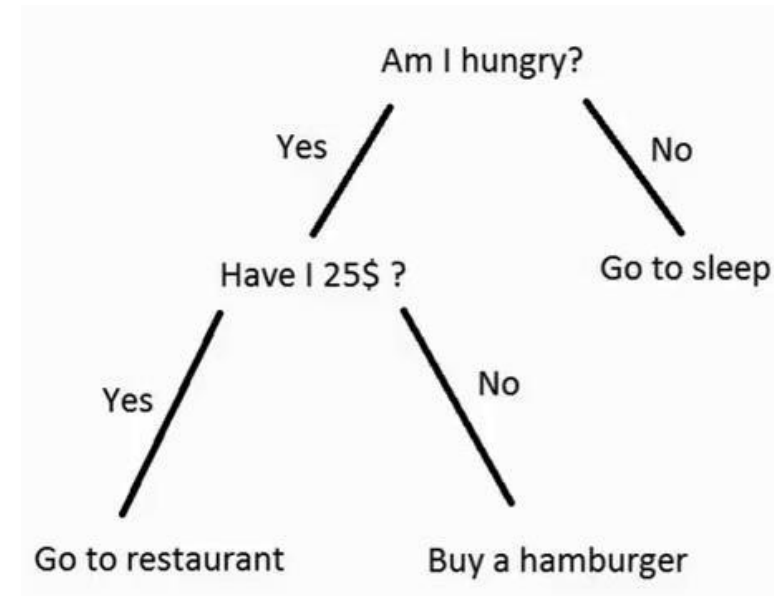
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Course Contents

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2. Agents and Environments
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5. Informed Search
6. Local Search
7. Constraint Satisfaction Problems
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11. Bayesian Networks, Naïve Bayes Classifier
12. Hidden Markov model, Markov Decision Process
- ➔ **13. Machine Learning (Supervised, Unsupervised, Reinforcement Learning)**

What is a decision tree?

- Decision tree is a graphical representation of all the possible solutions / decisions related to a given problem.
- Decision trees can be employed in “classification” and “regression” problems.
- Classification is usually performed using decision tree classifiers.



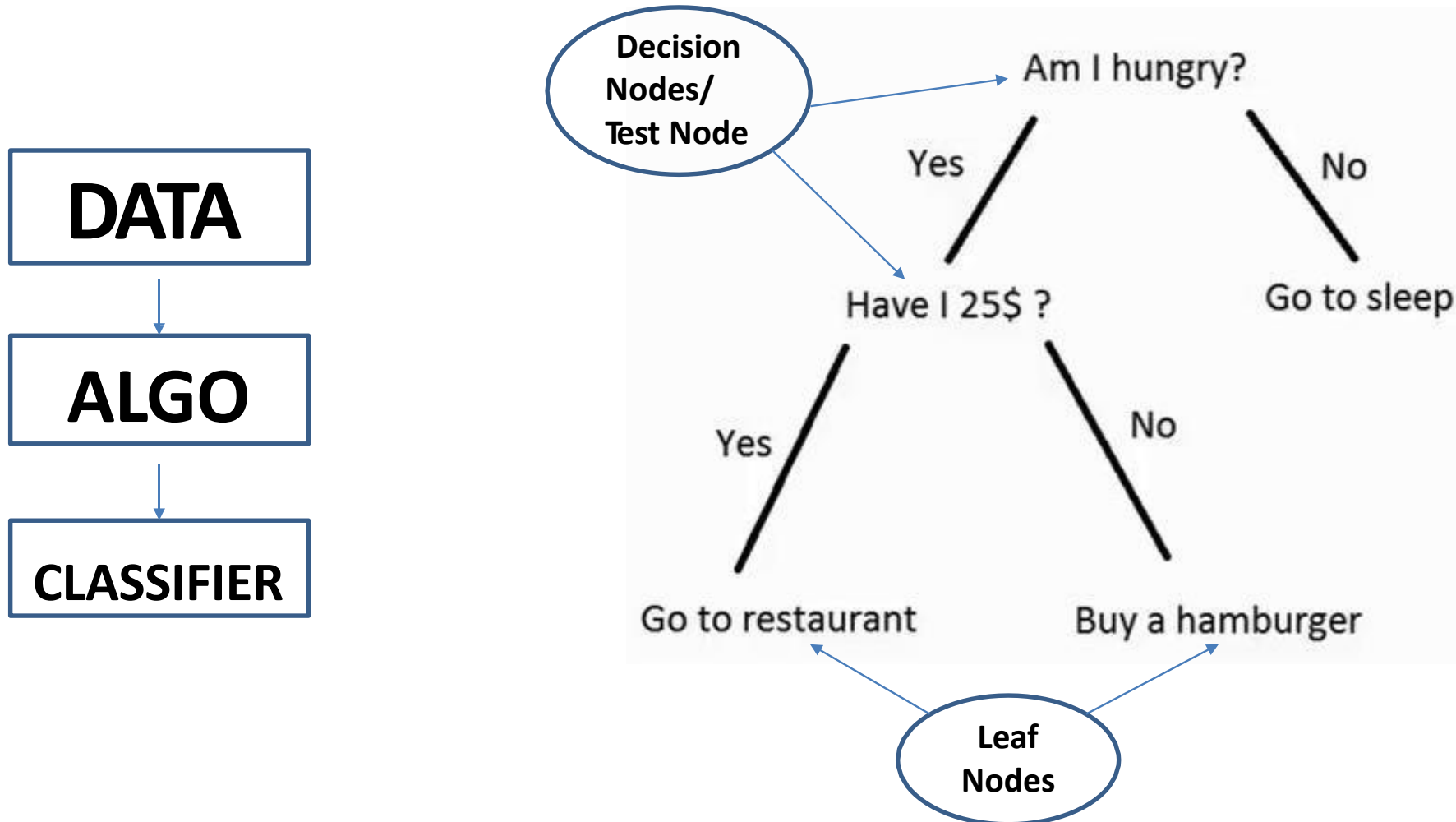
<https://www.youtube.com/watch?v=coOTec-0OGw>

<https://www.youtube.com/watch?v=JO2wiZif2OM>

<https://www.youtube.com/watch?v=2A8AGfxs0D8>

What is a decision tree?

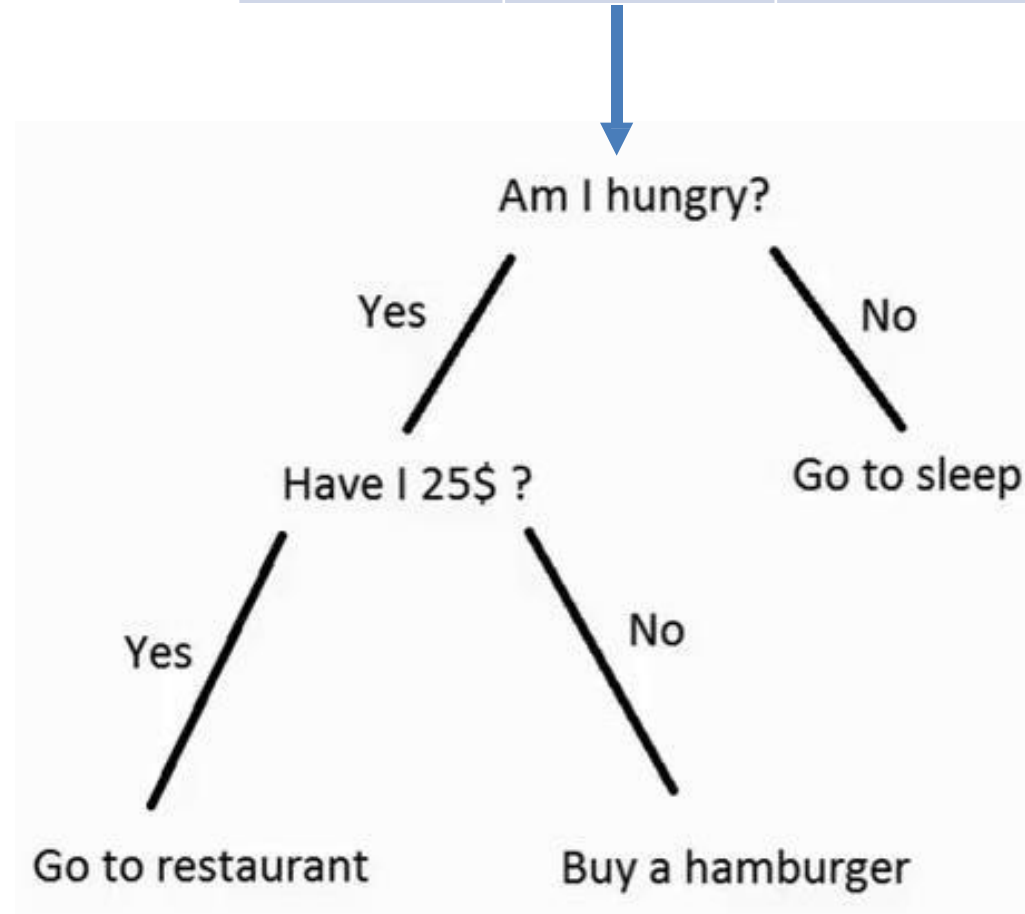
- It is a tree structures classifier



What is a decision tree?

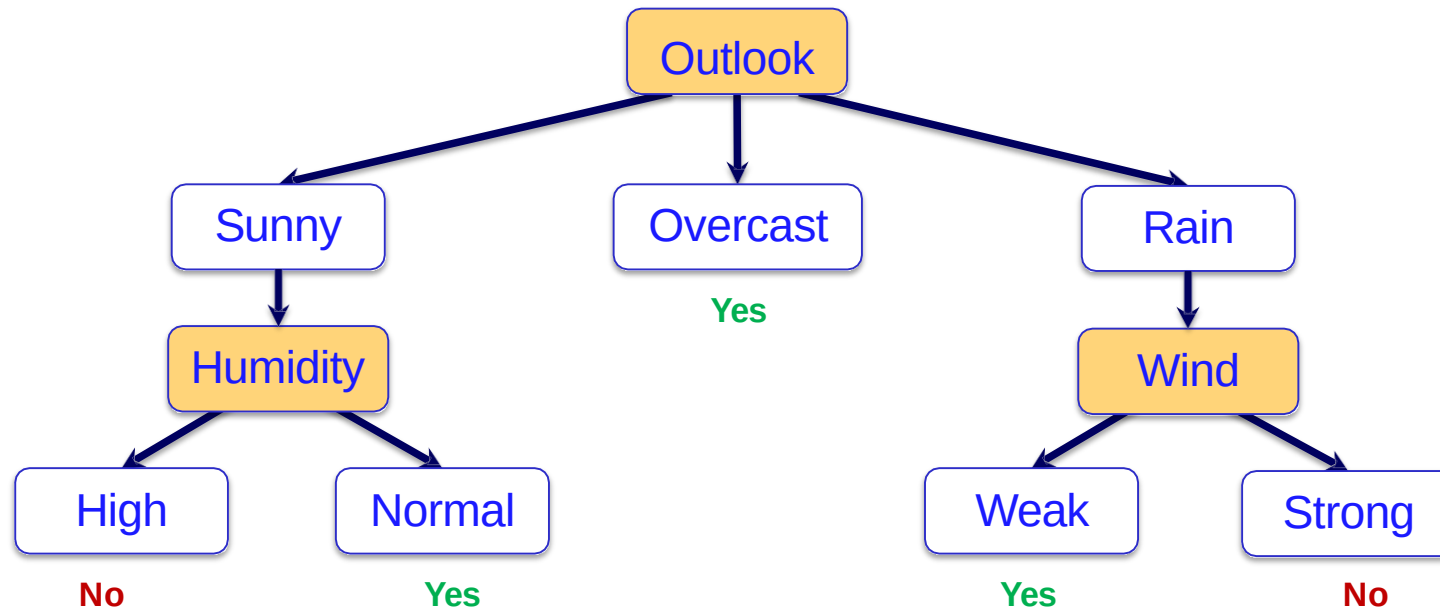
- Data Split
- Decision trees can be employed in “classification” and “regression” problems.
- Classification is usually performed using decision tree classifiers.

Am I Hungry	Have I \$25	Go to Sleep
Yes		
Yes		
No		
Yes		
No		



Example of a Decision Tree

Is it a good weather to play outside?



How to learn such a tree from past experience?

Decision Tree – Motivation Example

Given the following training examples, will you play in D15?

Divide and conquer:

- split into subsets
- are they pure?
(all yes or all no)
- if yes: stop
- if not: repeat

See which subset new data falls into

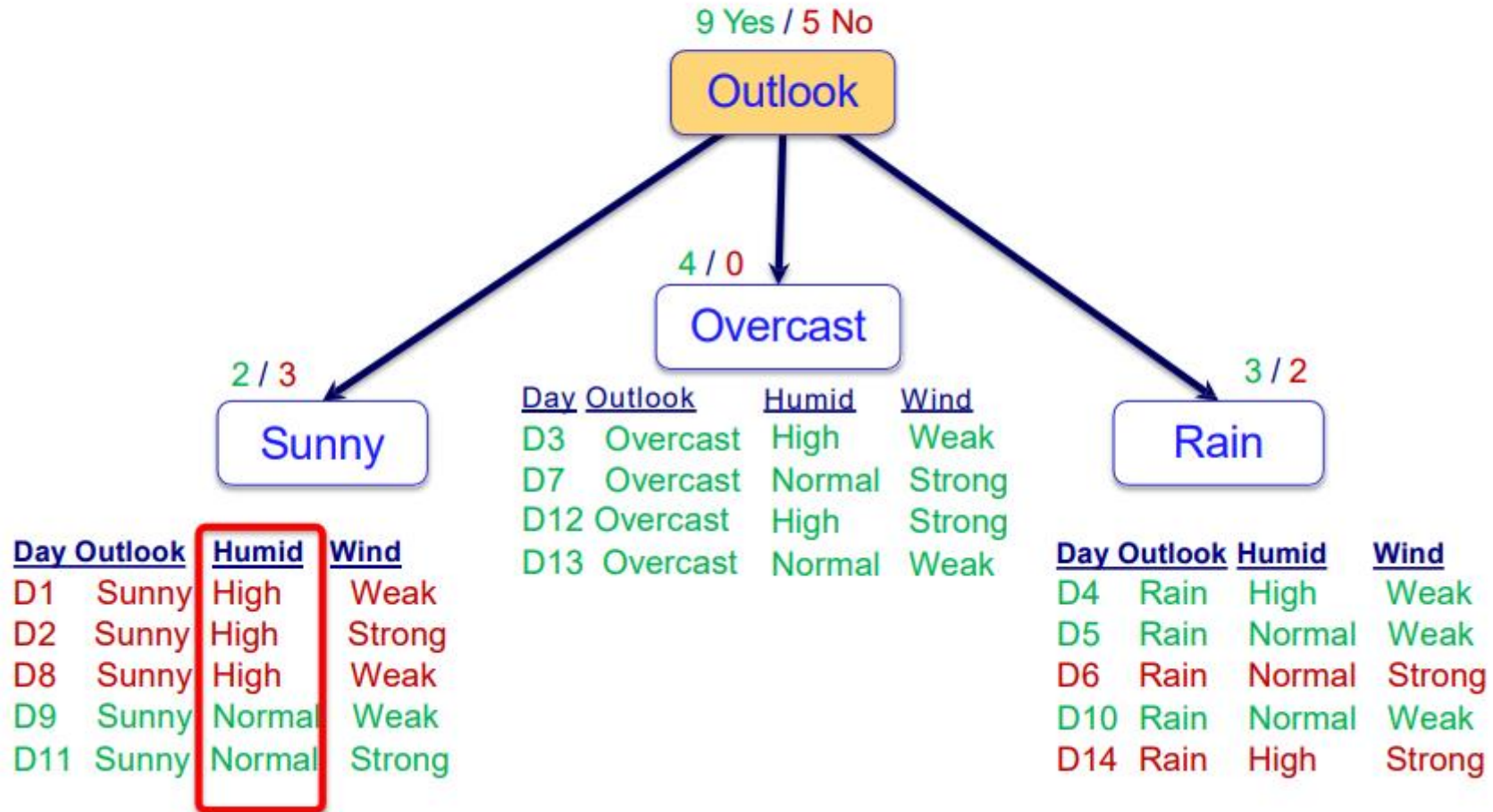
Training Examples

<u>Day</u>	<u>Outlook</u>	<u>Humidity</u>	<u>Wind</u>	<u>Play</u>
D1	Sunny	High	Weak	No
D2	Sunny	High	Strong	No
D3	Overcast	High	Weak	Yes
D4	Rain	High	Weak	Yes
D5	Rain	Normal	Weak	Yes
D6	Rain	Normal	Strong	No
D7	Overcast	Normal	Strong	Yes
D8	Sunny	High	Weak	No
D9	Sunny	Normal	Weak	Yes
D10	Rain	Normal	Weak	Yes
D11	Sunny	Normal	Strong	Yes
D12	Overcast	High	Strong	Yes
D13	Overcast	Normal	Weak	Yes
D14	Rain	High	Strong	No

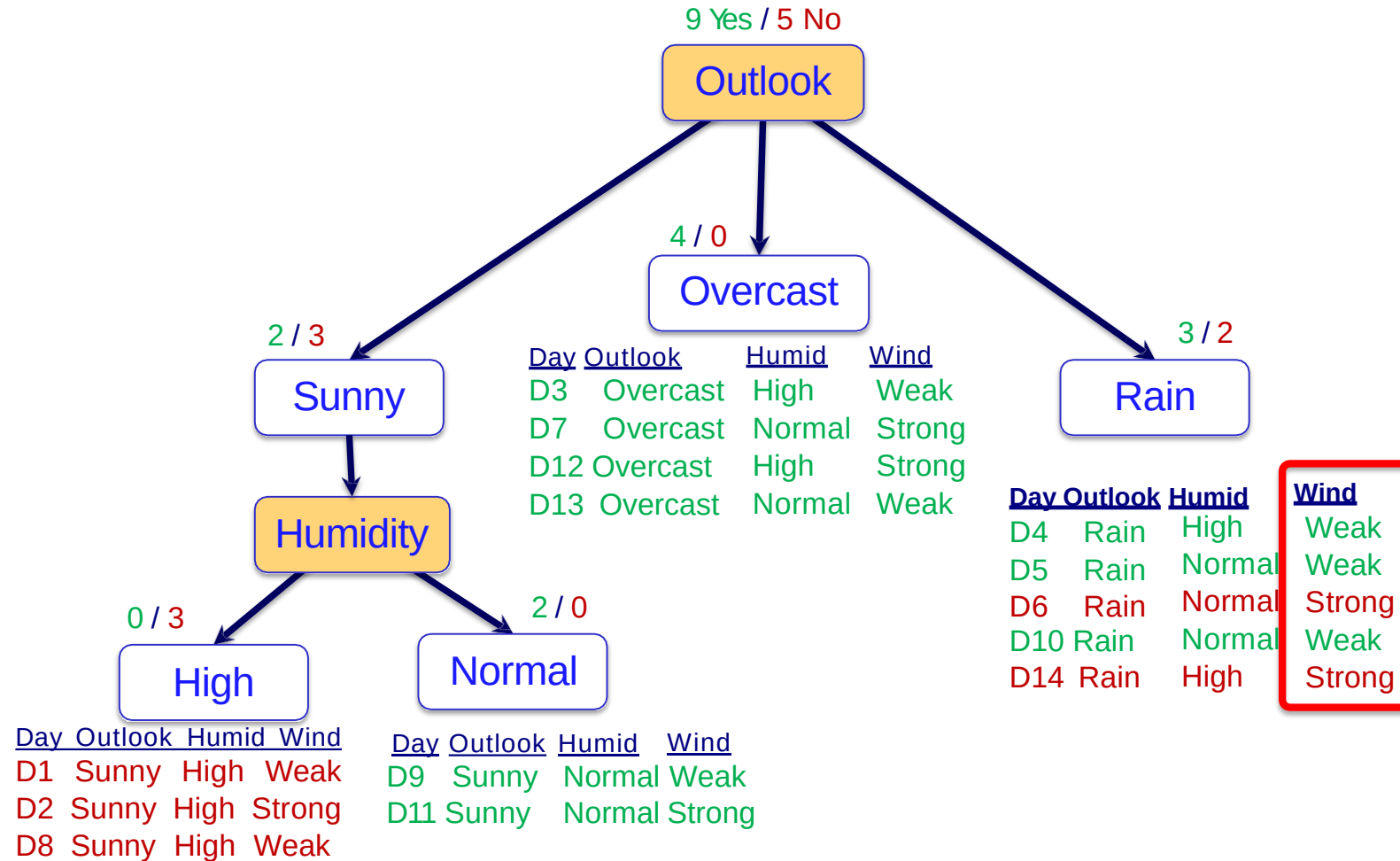
New data:

D15	Rain	High	Weak	???
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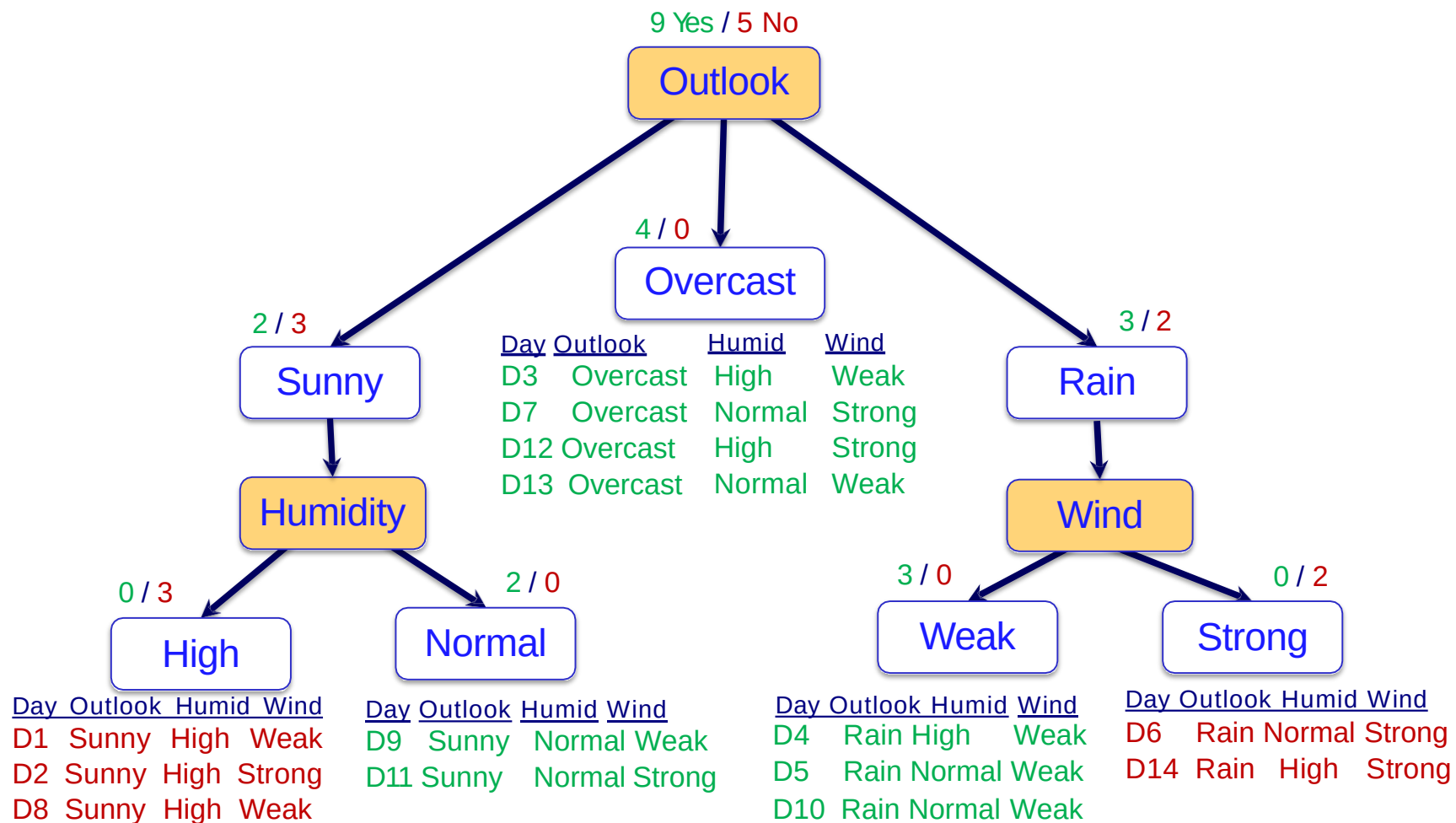
Decision Tree – Motivation Example



Decision Tree – Motivation Example

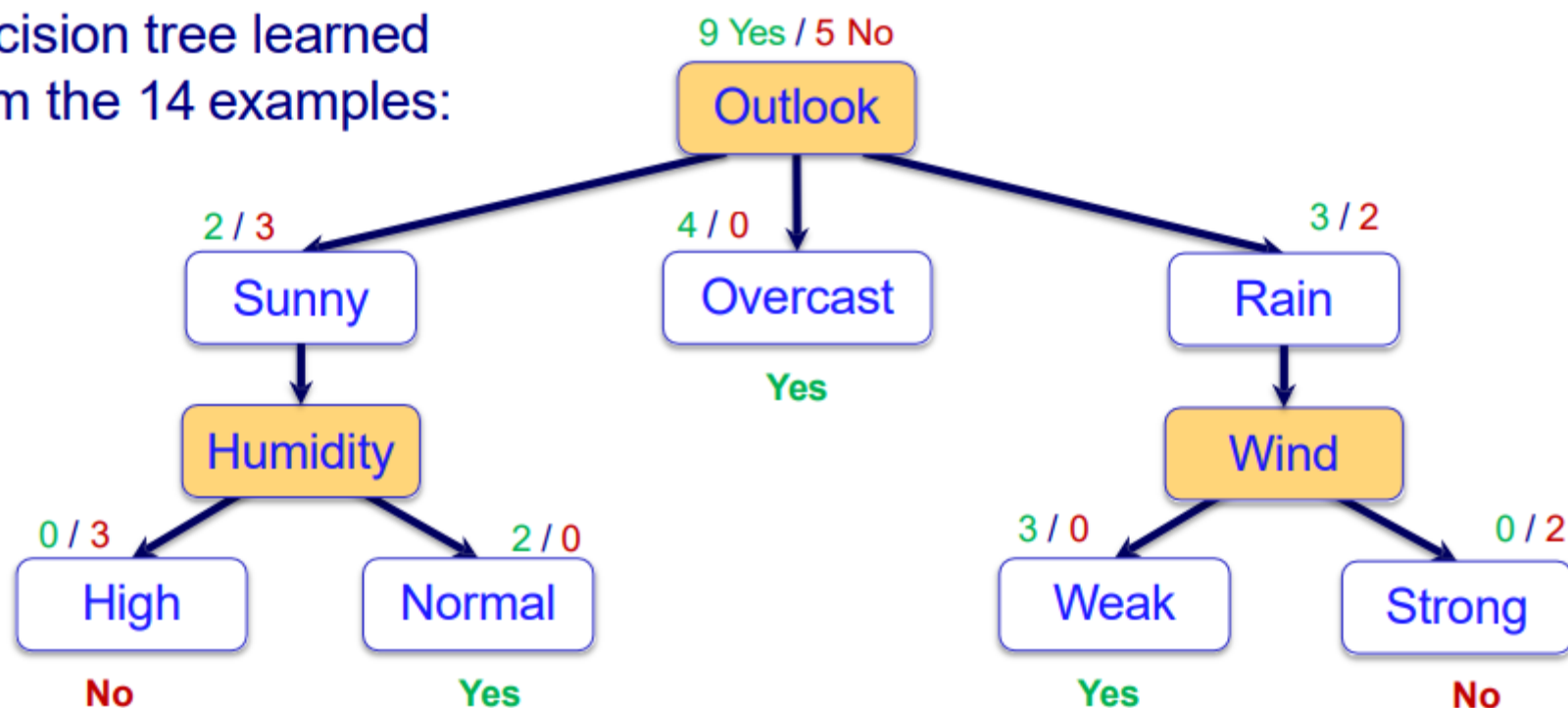


Decision Tree – Motivation Example



Decision Tree – Motivation Example

Decision tree learned
from the 14 examples:



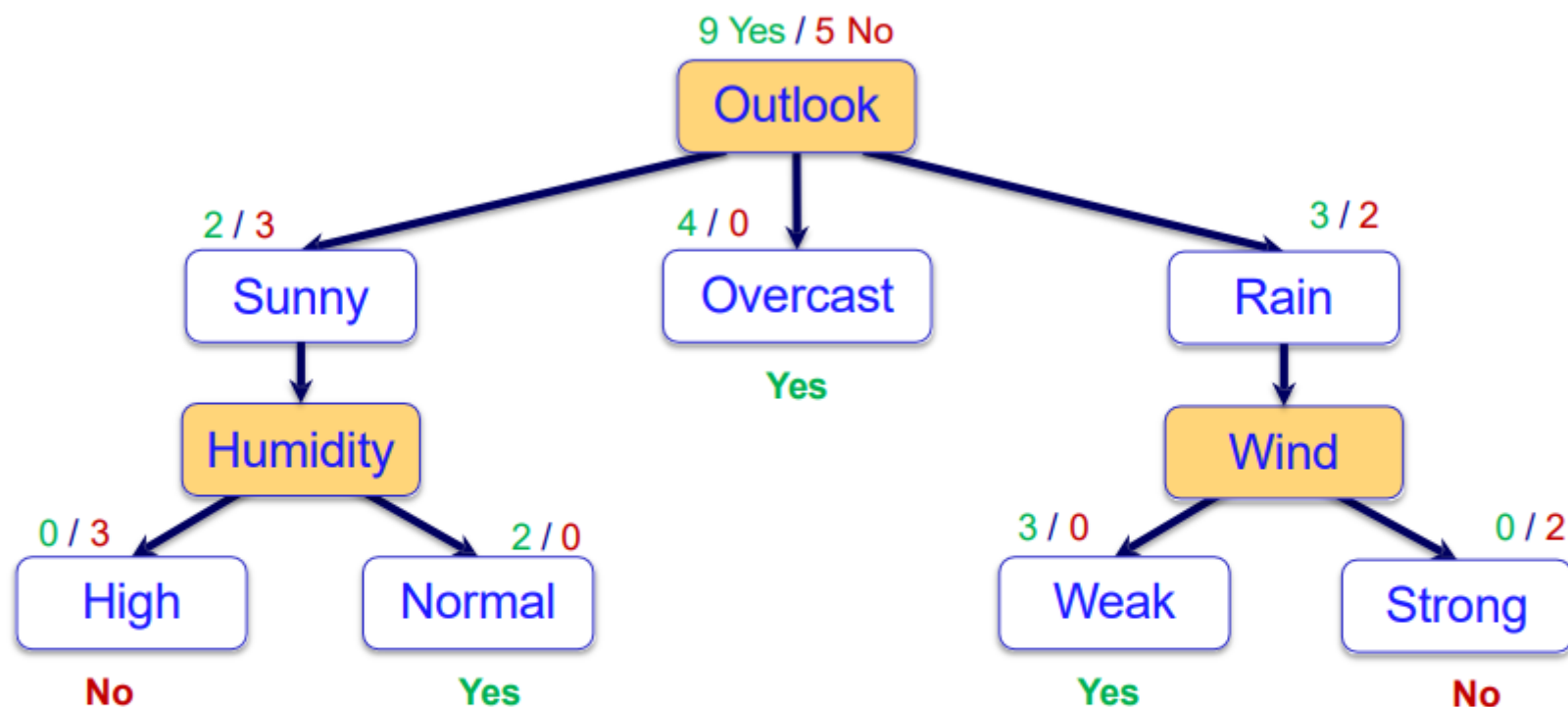
Decision Rule:

Yes $\Leftrightarrow$ (Outlook=Overcast) $\vee$
(Outlook=Sunny $\wedge$ Humidity=Normal) $\vee$
(Outlook=Rain $\wedge$ Wind=Weak)

Day	Outlook	Humid	Wind	
D15	Rain	High	Weak	???

→ Play

Decision Trees are Interpretable

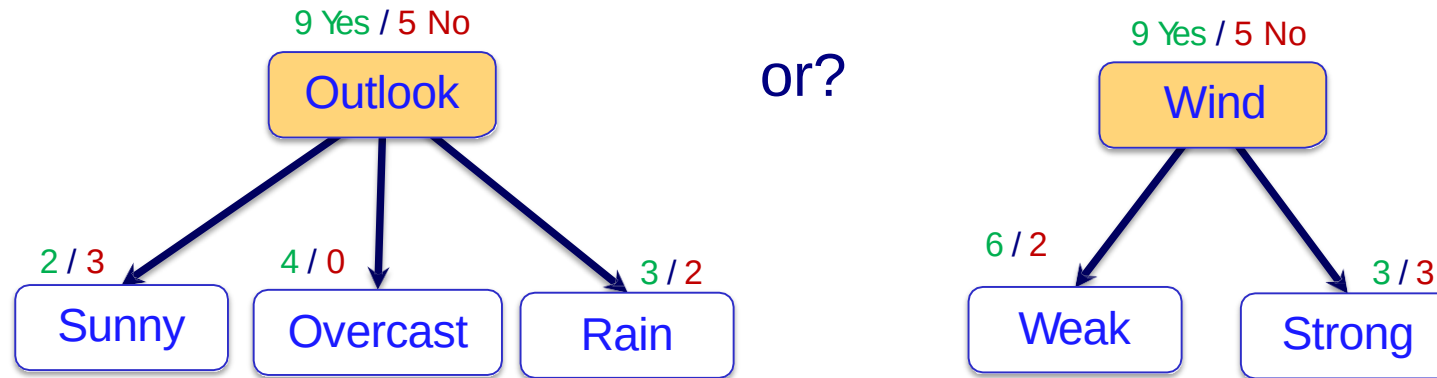


Disjunction of conjunctions of constraints on the attribute values of instances i.e., $(\dots \wedge \dots \wedge \dots) \vee (\dots \wedge \dots \wedge \dots) \vee \dots$

Set of **if-then-rules**, each branch represents one if-then-rule

- **if-part**: conjunctions of attribute tests on the nodes
- **then-part**: classification of the branch

Which attribute to split on?



Want to measure “purity” of the split

- more certain about Yes/No after the split
 - pure set (4 yes / 0 no) => completely certain (100%)
 - impure (3 yes / 3 no) => completely uncertain (50%)
- can’t use the probability of “yes” given the set, $P(\text{“yes”} \mid \text{set})$:
 - must be symmetric: 4 yes / 0 no as pure as 0 yes / 4 no

Entropy

Entropy tells us how much a set of data is pure/impure

For binary classification:

$$= -\frac{P}{P+N} \log_2\left(\frac{P}{P+N}\right) - \frac{N}{P+N} \log_2\left(\frac{N}{P+N}\right)$$

$$\text{Entropy}(S) = H(S) = -p_{\oplus} \log_2 p_{\oplus} - p_{\ominus} \log_2 p_{\ominus} \text{ bits}$$

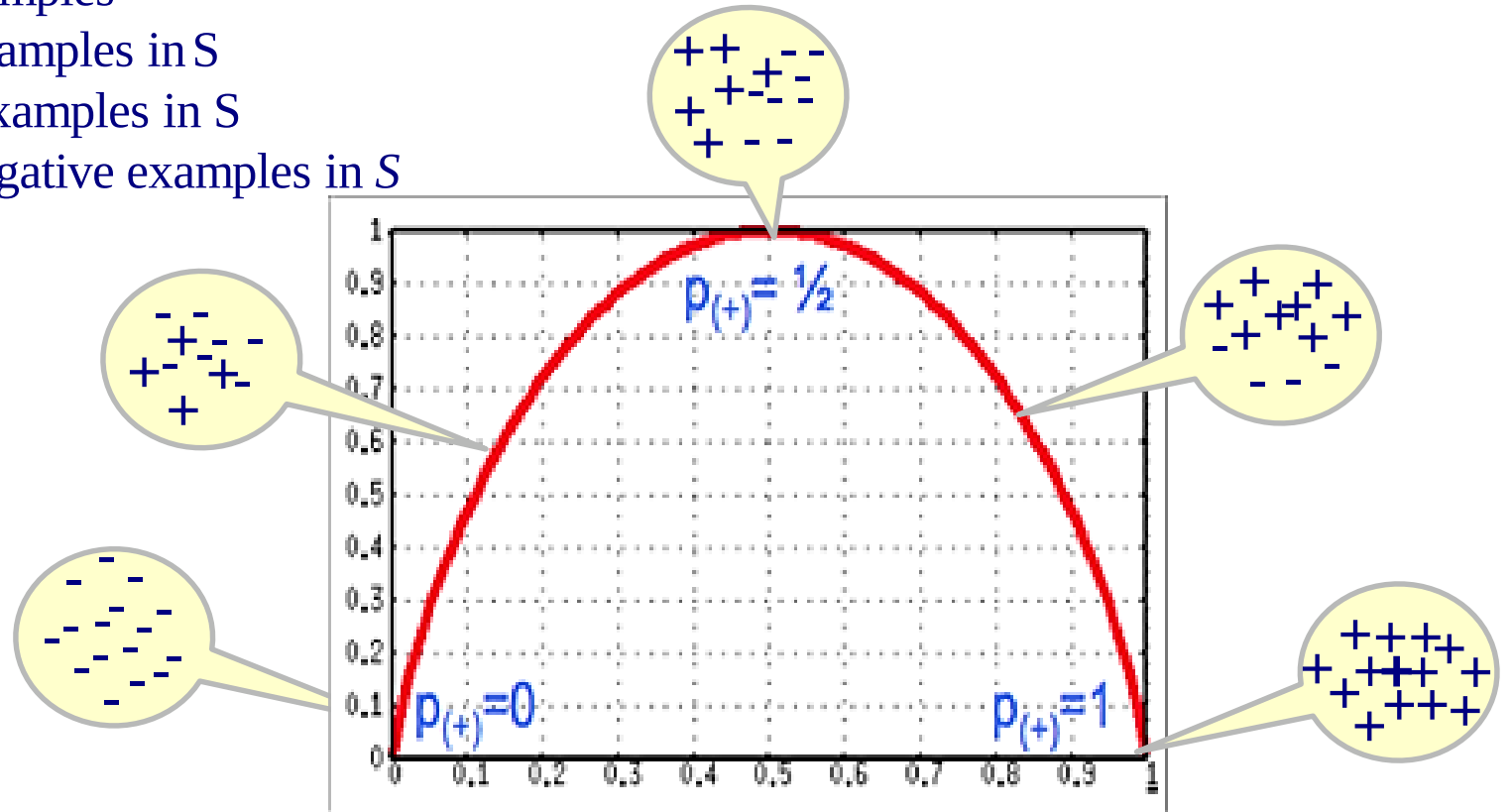
- S ... is a sample training examples
- $p_{\oplus}$ proportion of positive examples in S
- $p_{\ominus}$ proportion of negative examples in S
- $p_{\oplus} / p_{\ominus}$... % of positive/negative examples in S

- impure (3 yes / 3 no):

$$H(S) = -\frac{3}{6} \log_2 \frac{3}{6} - \frac{3}{6} \log_2 \frac{3}{6} = 1 \text{ bits}$$

- pure set (4 yes / 0 no):

$$H(S) = -\frac{4}{4} \log_2 \frac{4}{4} - \frac{0}{4} \log_2 \frac{0}{4} = 0 \text{ bits}$$



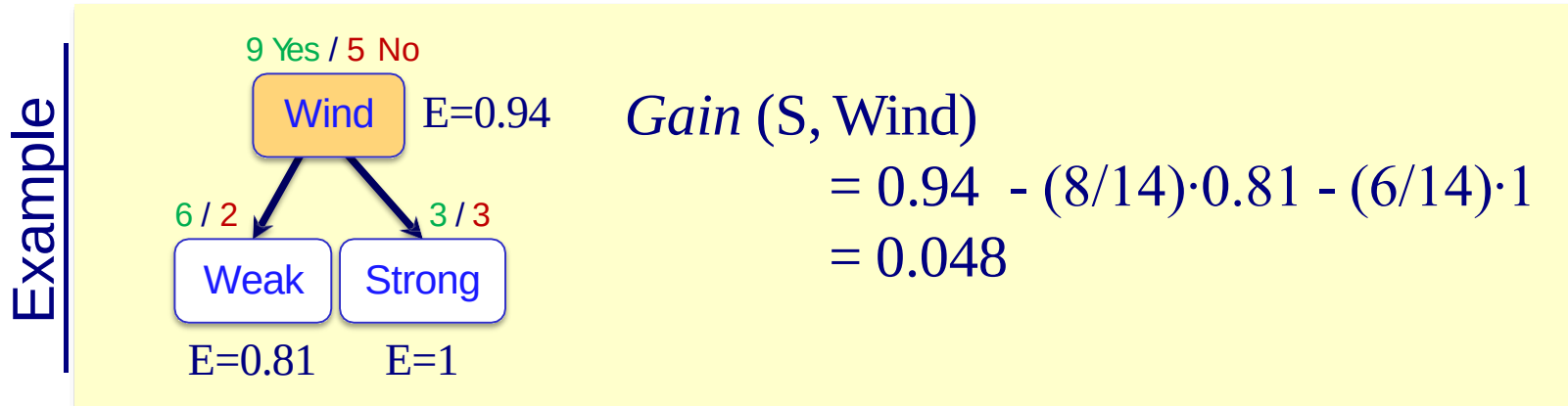
Information Gain

Entropy measures purity at each node, information gain looks at all nodes together and the expected drop in entropy after split.

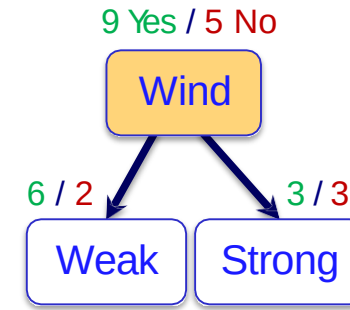
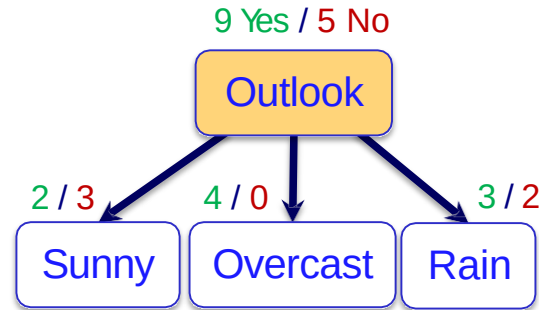
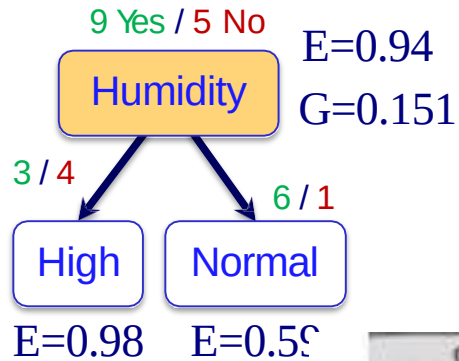
$\text{Gain}(S, A)$ = expected reduction in entropy due to sorting on A

$$\text{Gain}(S, A) = \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{S} \cdot \text{Entropy}(S_v)$$

Maximum $\text{Gain}(S, A)$ is selected!



Full Example (Entropy & Information Gain)



$$= -\frac{P}{P+N} \log_2\left(\frac{P}{P+N}\right) - \frac{N}{P+N} \log_2\left(\frac{N}{P+N}\right)$$

$$\text{Entropy}(\text{Humidity}) = -9/14 \cdot \log_2(9/14) - 5/14 \cdot \log_2(5/14) = 0.94$$

$$\text{Entropy}(\text{High}) = -3/7 \cdot \log_2(3/7) - 4/7 \cdot \log_2(4/7) = 0.98$$

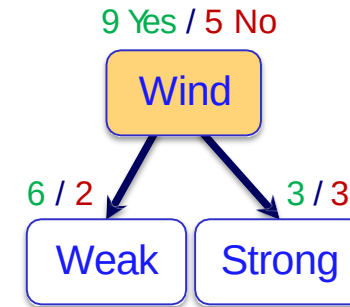
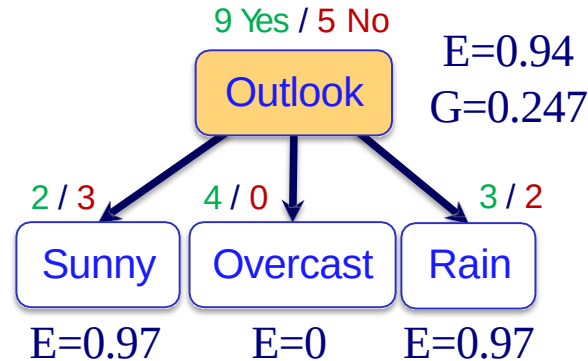
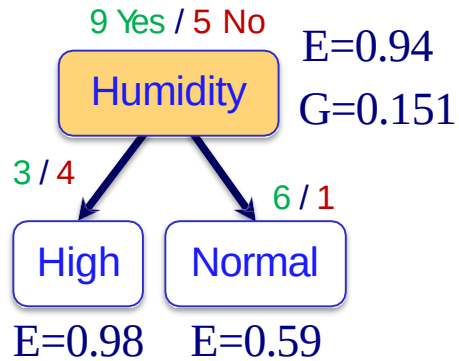
$$\text{Entropy}(\text{Normal}) = -6/7 \cdot \log_2(6/7) - 1/7 \cdot \log_2(1/7) = 0.59$$

$$\text{Gain}(S, A) = \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{S} \cdot \text{Entropy}(S_v)$$

$$\text{Gain}(S, \text{Humidity}) = 0.94 - (7/14) \cdot 0.98 - (7/14) \cdot 0.59 = 0.151$$

Day	Outlook	Humidity	Wind	Play
D1	Sunny	High	Weak	No
D2	Sunny	High	Strong	No
D3	Overcast	High	Weak	Yes
D4	Rain	High	Weak	Yes
D5	Rain	Normal	Weak	Yes
D6	Rain	Normal	Strong	No
D7	Overcast	Normal	Strong	Yes
D8	Sunny	High	Weak	No
D9	Sunny	Normal	Weak	Yes
D10	Rain	Normal	Weak	Yes
D11	Sunny	Normal	Strong	Yes
D12	Overcast	High	Strong	Yes
D13	Overcast	Normal	Weak	Yes
D14	Rain	High	Strong	No

Full Example (Entropy & Information Gain)



$$Entropy(Outlook) = -9/14 \cdot \log_2(9/14) - 5/14 \cdot \log_2(5/14) = 0.94$$

$$Entropy(Sunny) = -2/5 \cdot \log_2(2/5) - 3/5 \cdot \log_2(3/5) = 0.97$$

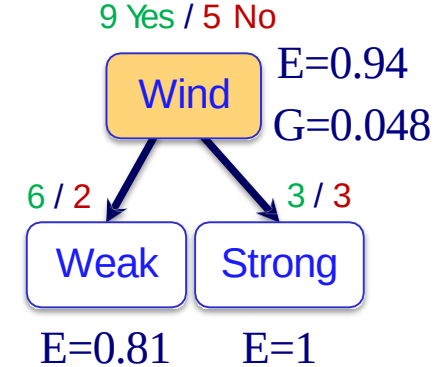
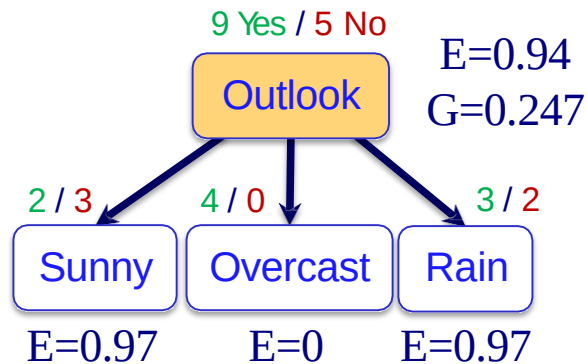
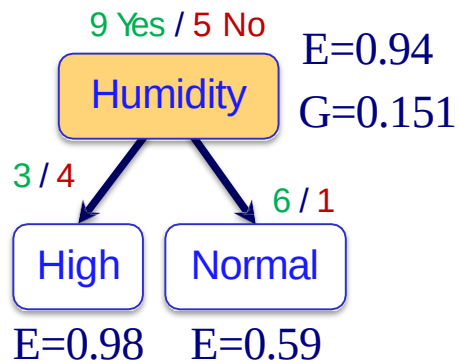
$$Entropy(Overcast) = -4/4 \cdot \log_2(4/4) - 0/4 \cdot \log_2(0/4) = 0$$

$$Entropy(Rain) = -3/5 \cdot \log_2(3/5) - 2/5 \cdot \log_2(2/5) = 0.97$$

$$Gain(S, Outlook) = 0.94 - (5/14) \cdot 0.97 - (4/14) \cdot 0 - (5/14) \cdot 0.97 = 0.247$$

Day	Outlook	Humidity	Wind	Play
D1	Sunny	High	Weak	No
D2	Sunny	High	Strong	No
D3	Overcast	High	Weak	Yes
D4	Rain	High	Weak	Yes
D5	Rain	Normal	Weak	Yes
D6	Rain	Normal	Strong	No
D7	Overcast	Normal	Strong	Yes
D8	Sunny	High	Weak	No
D9	Sunny	Normal	Weak	Yes
D10	Rain	Normal	Weak	Yes
D11	Sunny	Normal	Strong	Yes
D12	Overcast	High	Strong	Yes
D13	Overcast	Normal	Weak	Yes
D14	Rain	High	Strong	No

Full Example (Entropy & Information Gain)



$$\text{Entropy (Wind)} = -9/14 \cdot \log_2(9/14) - 5/14 \cdot \log_2(5/14) = 0.94$$

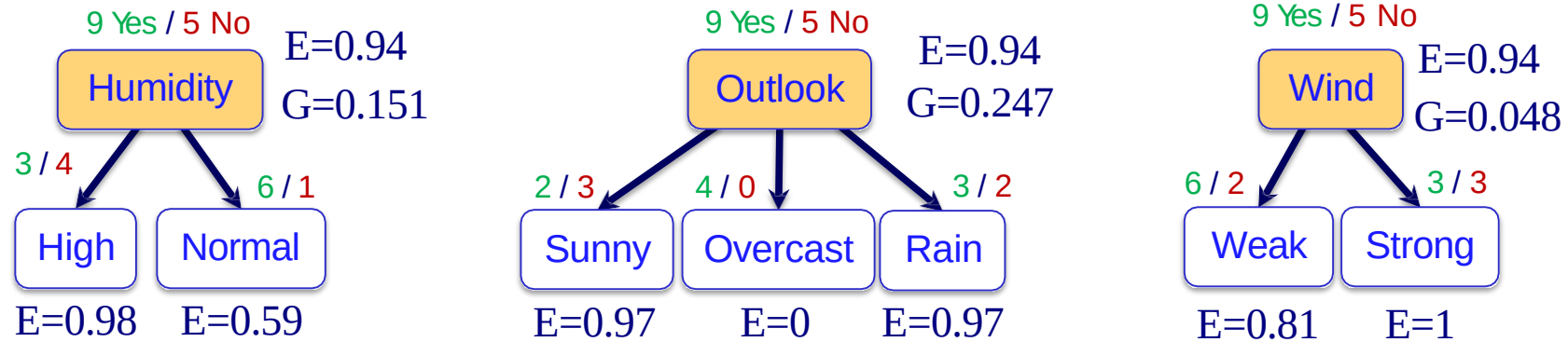
$$\text{Entropy (Weak)} = -6/8 \cdot \log_2(6/8) - 2/8 \cdot \log_2(2/8) = 0.81$$

$$\text{Entropy (Strong)} = -3/6 \cdot \log_2(3/6) - 3/6 \cdot \log_2(3/6) = 1$$

$$\text{Gain (S, Wind)} = 0.94 - (8/14) \cdot 0.81 - (6/14) \cdot 1 = 0.048$$

Day	Outlook	Humidity	Wind	Play
D1	Sunny	High	Weak	No
D2	Sunny	High	Strong	No
D3	Overcast	High	Weak	Yes
D4	Rain	High	Weak	Yes
D5	Rain	Normal	Weak	Yes
D6	Rain	Normal	Strong	No
D7	Overcast	Normal	Strong	Yes
D8	Sunny	High	Weak	No
D9	Sunny	Normal	Weak	Yes
D10	Rain	Normal	Weak	Yes
D11	Sunny	Normal	Strong	Yes
D12	Overcast	High	Strong	Yes
D13	Overcast	Normal	Weak	Yes
D14	Rain	High	Strong	No

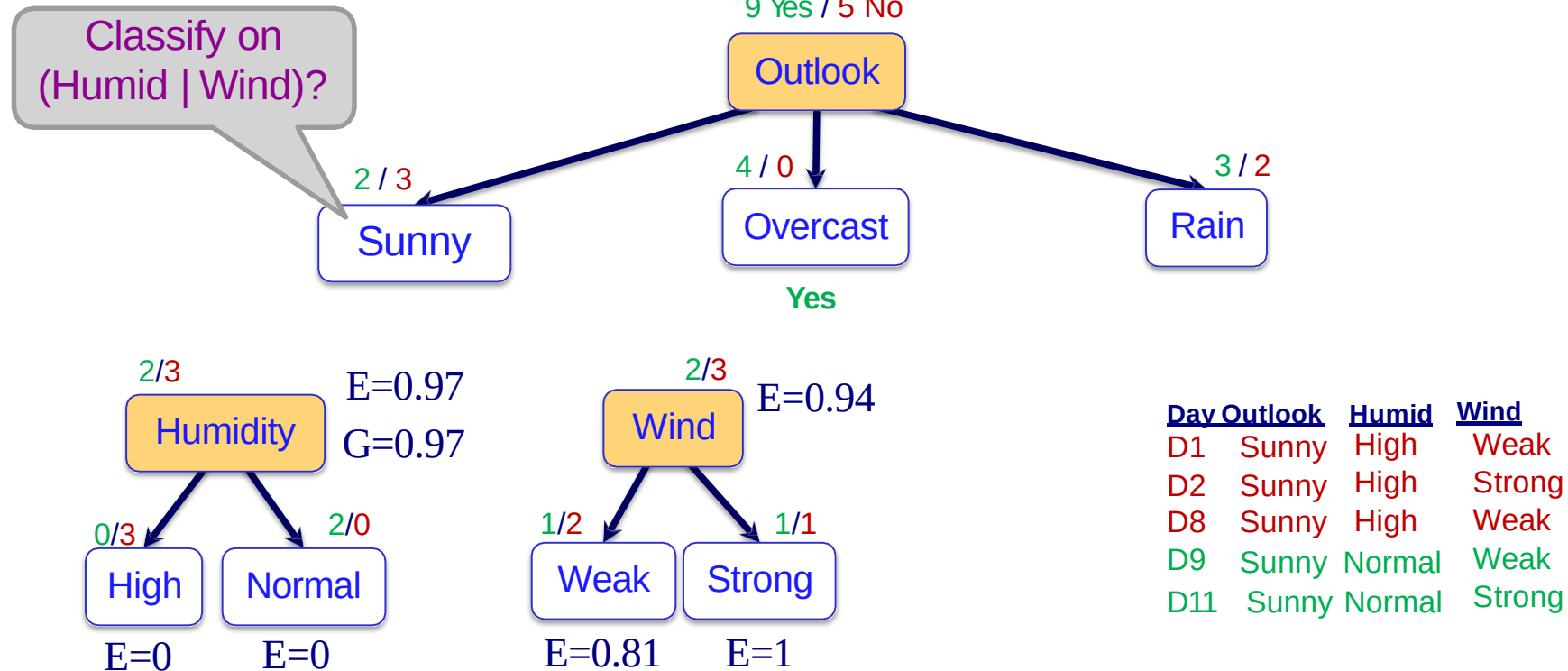
Full Example (Entropy & Information Gain)



The attribute with the largest Information Gain (Outlook 0.247) is selected as the decision node.

Nodes with zero Entropy (e.g., Overcast) does not need splitting

Full Example (Entropy & Information Gain)



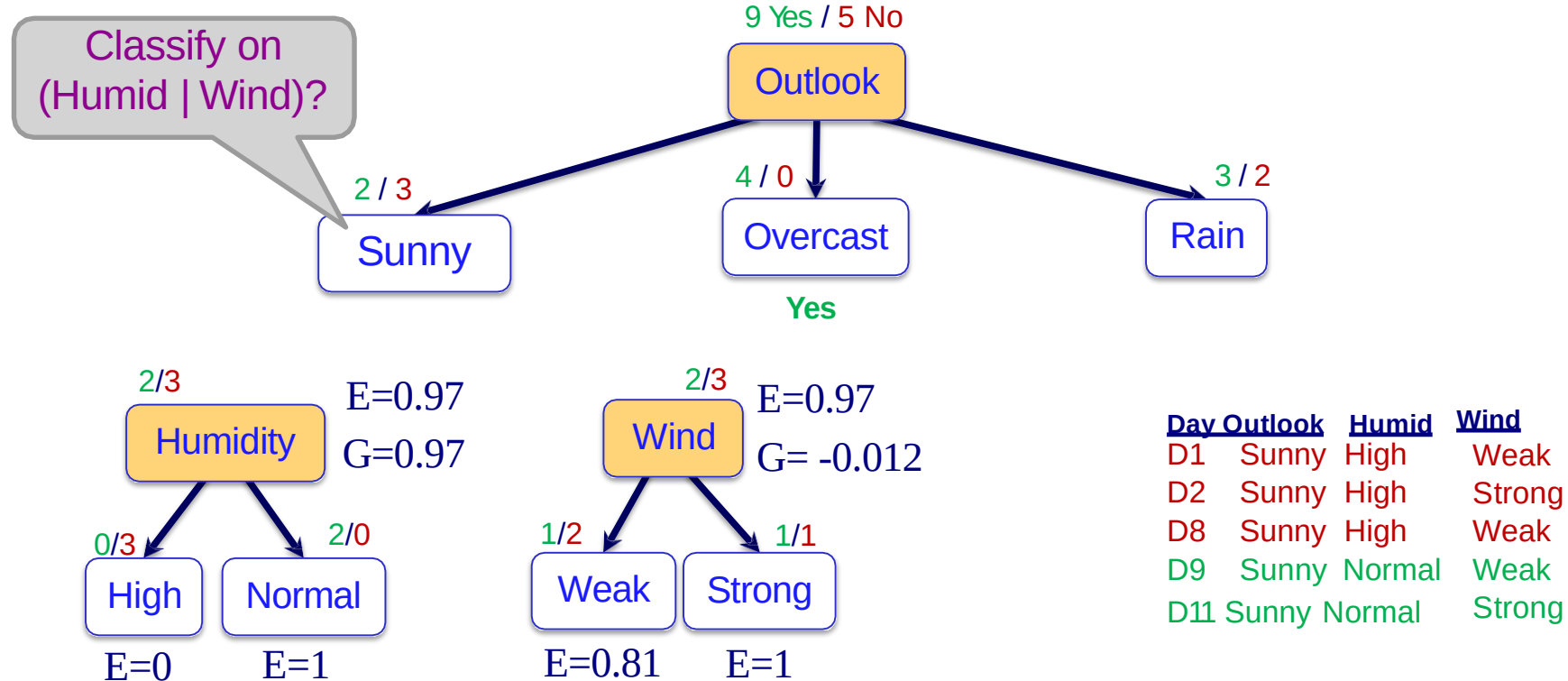
$$Entropy(\text{Humidity}) = -2/5 \cdot \log_2(2/5) - 3/5 \cdot \log_2(3/5) = 0.97$$

$$Entropy(\text{High}) = -0/3 \cdot \log_2(0/3) - 3/3 \cdot \log_2(3/3) = 0$$

$$Entropy(\text{Normal}) = -2/2 \cdot \log_2(2/2) - 0/2 \cdot \log_2(0/2) = 0$$

$$Gain(S, \text{Humidity}) = 0.97 - (3/5) \cdot 0 - (2/5) \cdot 0 = 0.97$$

Full Example (Entropy & Information Gain)



$$\text{Entropy (Wind)} = -2/5 \cdot \log_2(2/5) - 3/5 \cdot \log_2(3/5) = 0.97$$

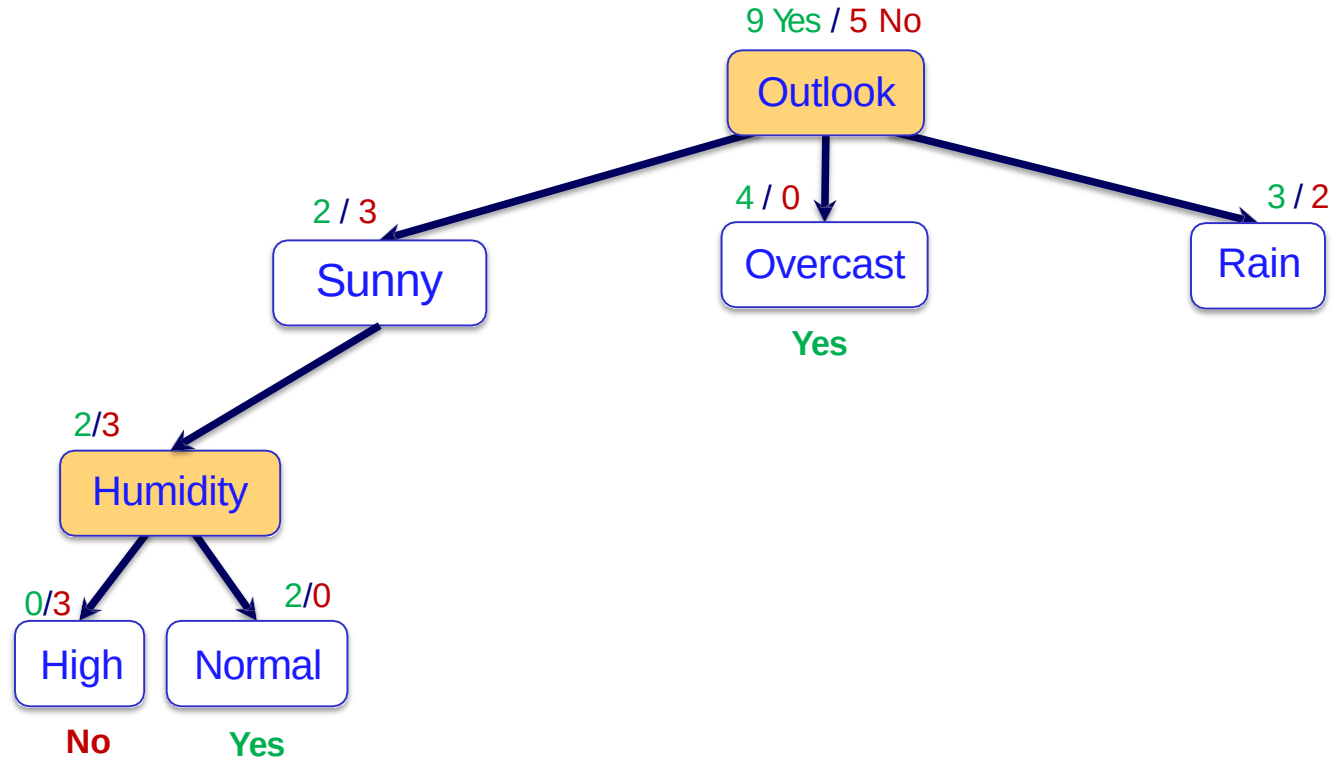
$$\text{Entropy (Weak)} = -1/3 \cdot \log_2(1/3) - 2/3 \cdot \log_2(2/3) = 0.92$$

$$\text{Entropy (Strong)} = -1/2 \cdot \log_2(1/2) - 1/2 \cdot \log_2(1/2) = 1$$

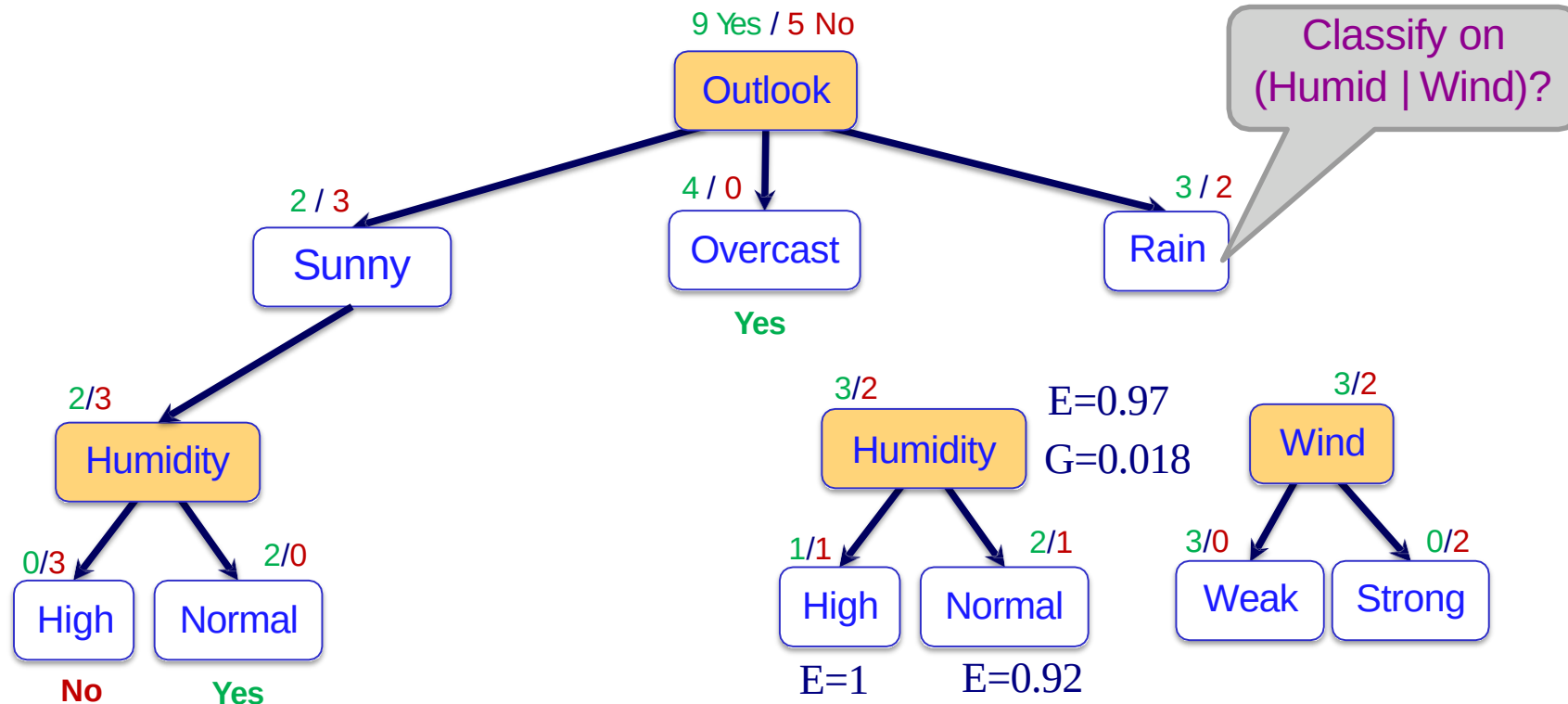
$$\text{Gain (S, Wind)} = 0.97 - (3/5) \cdot 0.97 - (2/5) \cdot 1 = -0.012$$

→ Humidity has the highest gain (0.97)

Full Example (Entropy & Information Gain)



Full Example (Entropy & Information Gain)



$$Entropy(Humidity) = -3/5 \cdot \log_2(3/5) - 2/5 \cdot \log_2(2/5) = 0.97$$

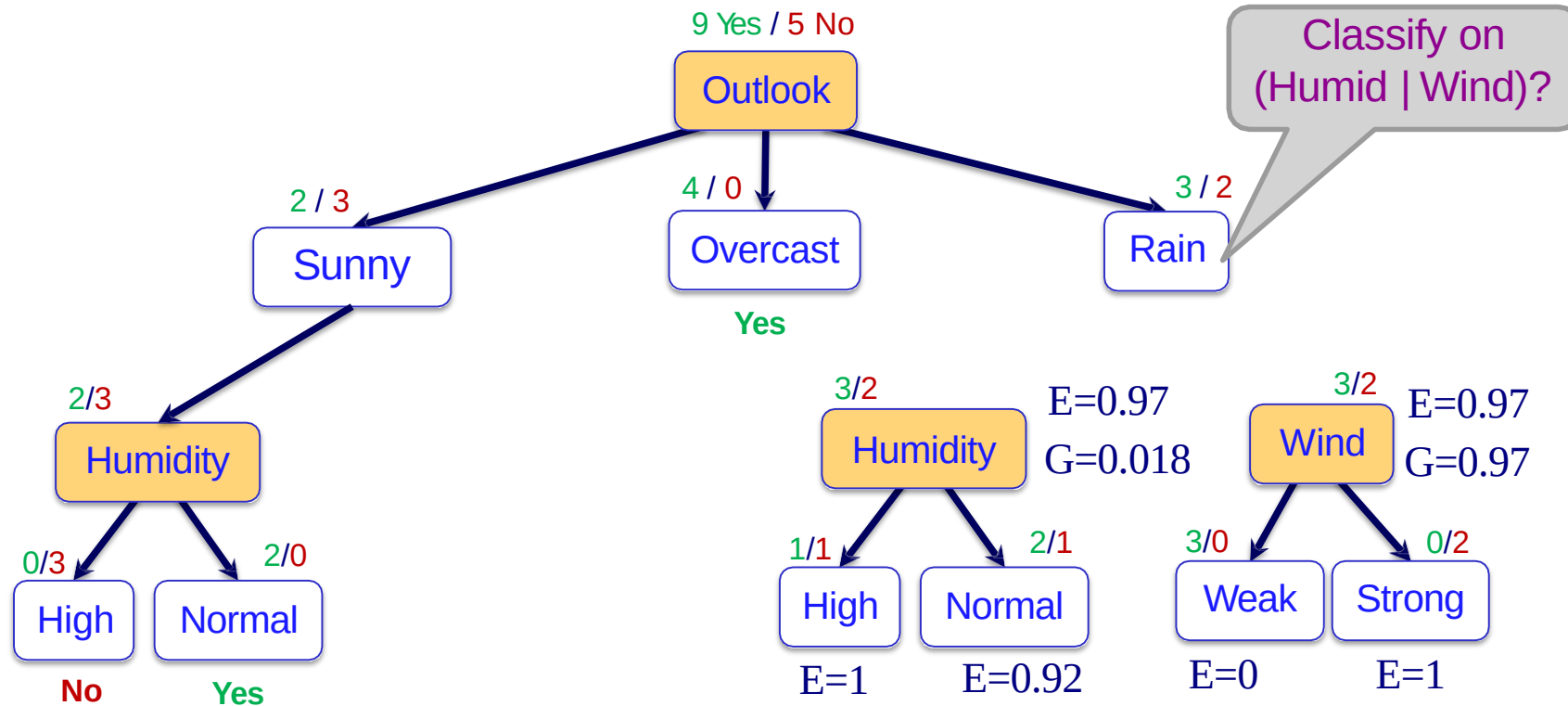
$$Entropy(High) = -1/2 \cdot \log_2(1/2) - 1/2 \cdot \log_2(1/2) = 1$$

$$Entropy(Normal) = -2/3 \cdot \log_2(2/3) - 1/3 \cdot \log_2(1/3) = 0.92$$

$$Gain(S, Humidity) = 0.97 - (2/5) \cdot 1 - (3/5) \cdot 0.92 = 0.018$$

Day	Outlook	Humid	Wind
D4	Rain	High	Weak
D5	Rain	Normal	Weak
D6	Rain	Normal	Strong
D10	Rain	Normal	Weak
D14	Rain	High	Strong

Full Example (Entropy & Information Gain)



$$Entropy(Wind) = -3/5 \cdot \log_2(3/5) - 2/5 \cdot \log_2(2/5) = 0.97$$

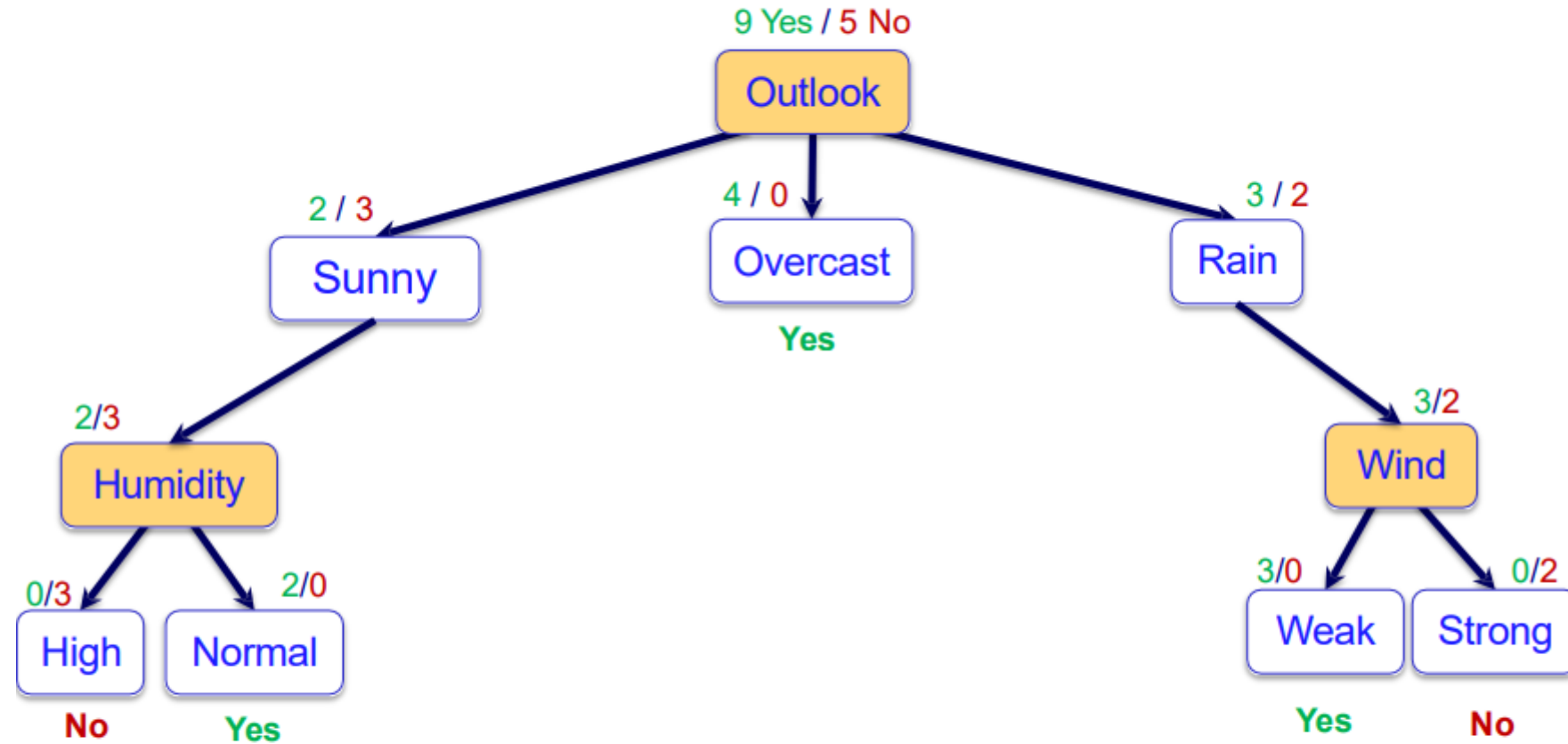
$$Entropy(Weak) = -3/3 \cdot \log_2(3/3) - 0/3 \cdot \log_2(0/3) = 0$$

$$Entropy(Strong) = -0/2 \cdot \log_2(0/2) - 2/2 \cdot \log_2(2/2) = 0$$

$$Gain(S, Wind) = 0.97 - (3/5) \cdot 0 - (2/5) \cdot 0 = 0.97$$

Day	Outlook	Humid	Wind
D4	Rain	High	Weak
D5	Rain	Normal	Weak
D6	Rain	Normal	Strong
D10	Rain	Normal	Weak
D14	Rain	High	Strong

Full Example (Entropy & Information Gain)



Decision Rule:

Yes $\Leftrightarrow$ (Outlook=Overcast) $\vee$
(Outlook=Sunny $\wedge$ Humidity=Normal) $\vee$
(Outlook=Rain $\wedge$ Wind=Weak)

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Attribute: Outlook

Values (Outlook) = Sunny, Overcast, Rain

$S = [9+, 5-]$

$S_{Sunny} \leftarrow [2+, 3-]$

$S_{Overcast} \leftarrow [4+, 0-]$

$S_{Rain} \leftarrow [3+, 2-]$

Entropy = $-\frac{P}{P+N} \log_2\left(\frac{P}{P+N}\right) - \frac{N}{P+N} \log_2\left(\frac{N}{P+N}\right)$

$Entropy(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$

$Entropy(S_{Sunny}) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.971$

$Entropy(S_{Overcast}) = -\frac{4}{4} \log_2 \frac{4}{4} - \frac{0}{4} \log_2 \frac{0}{4} = 0$

$Entropy(S_{Rain}) = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.971$

$Gain(S, Outlook) = Entropy(S) - \sum_{v \in \{Sunny, Overcast, Rain\}} \frac{|S_v|}{|S|} Entropy(S_v)$

$Gain(S, Outlook)$

$= Entropy(S) - \frac{5}{14} Entropy(S_{Sunny}) - \frac{4}{14} Entropy(S_{Overcast}) - \frac{5}{14} Entropy(S_{Rain})$

$Gain(S, Outlook) = 0.94 - \frac{5}{14} 0.971 - \frac{4}{14} 0 - \frac{5}{14} 0.971 = 0.2464$

$$\text{Entropy} = -\frac{P}{P+N} \log_2\left(\frac{P}{P+N}\right) - \frac{N}{P+N} \log_2\left(\frac{N}{P+N}\right)$$

Attribute: Temp

Values (Temp) = Hot, Mild, Cool

$$S = [9+, 5-]$$

$$\text{Entropy}(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$$

$$S_{\text{Hot}} \leftarrow [2+, 2-]$$

$$\text{Entropy}(S_{\text{Hot}}) = -\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4} = 1.0$$

$$S_{\text{Mild}} \leftarrow [4+, 2-]$$

$$\text{Entropy}(S_{\text{Mild}}) = -\frac{4}{6} \log_2 \frac{4}{6} - \frac{2}{6} \log_2 \frac{2}{6} = 0.9183$$

$$S_{\text{Cool}} \leftarrow [3+, 1-]$$

$$\text{Entropy}(S_{\text{Cool}}) = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} = 0.8113$$

$$\text{Gain}(S, \text{Temp}) = \text{Entropy}(S) - \sum_{v \in \{\text{Hot}, \text{Mild}, \text{Cool}\}} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

$$\text{Gain}(S, \text{Temp})$$

$$= \text{Entropy}(S) - \frac{4}{14} \text{Entropy}(S_{\text{Hot}}) - \frac{6}{14} \text{Entropy}(S_{\text{Mild}})$$

$$- \frac{4}{14} \text{Entropy}(S_{\text{Cool}})$$

$$\text{Gain}(S, \text{Temp}) = 0.94 - \frac{4}{14} 1.0 - \frac{6}{14} 0.9183 - \frac{4}{14} 0.8113 = 0.0289$$

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Attribute: Humidity

$$\text{Entropy} = -\frac{P}{P+N} \log_2\left(\frac{P}{P+N}\right) - \frac{N}{P+N} \log_2\left(\frac{N}{P+N}\right)$$

Values (Humidity) = High, Normal

$$S = [9+, 5-]$$

$$\text{Entropy}(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$$

$$S_{\text{High}} \leftarrow [3+, 4-]$$

$$\text{Entropy}(S_{\text{High}}) = -\frac{3}{7} \log_2 \frac{3}{7} - \frac{4}{7} \log_2 \frac{4}{7} = 0.9852$$

$$S_{\text{Normal}} \leftarrow [6+, 1-]$$

$$\text{Entropy}(S_{\text{Normal}}) = -\frac{6}{7} \log_2 \frac{6}{7} - \frac{1}{7} \log_2 \frac{1}{7} = 0.5916$$

$$\text{Gain}(S, \text{Humidity}) = \text{Entropy}(S) - \sum_{v \in \{\text{High}, \text{Normal}\}} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

$$\text{Gain}(S, \text{Humidity})$$

$$= \text{Entropy}(S) - \frac{7}{14} \text{Entropy}(S_{\text{High}}) - \frac{7}{14} \text{Entropy}(S_{\text{Normal}})$$

$$\text{Gain}(S, \text{Humidity}) = 0.94 - \frac{7}{14} 0.9852 - \frac{7}{14} 0.5916 = 0.1516$$

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Attribute: Wind

$$\text{Entropy} = -\frac{P}{P+N} \log_2\left(\frac{P}{P+N}\right) - \frac{N}{P+N} \log_2\left(\frac{N}{P+N}\right)$$

Values (Wind) = Strong, Weak

$$S = [9+, 5-]$$

$$\text{Entropy}(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$$

$$S_{\text{Strong}} \leftarrow [3+, 3-]$$

$$\text{Entropy}(S_{\text{Strong}}) = 1.0$$

$$S_{\text{Weak}} \leftarrow [6+, 2-]$$

$$\text{Entropy}(S_{\text{Weak}}) = -\frac{6}{8} \log_2 \frac{6}{8} - \frac{2}{8} \log_2 \frac{2}{8} = 0.8113$$

$$\text{Gain}(S, \text{Wind}) = \text{Entropy}(S) - \sum_{v \in (\text{Strong}, \text{Weak})} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

$$\text{Gain}(S, \text{Wind}) = \text{Entropy}(S) - \frac{6}{14} \text{Entropy}(S_{\text{Strong}}) - \frac{8}{14} \text{Entropy}(S_{\text{Weak}})$$

↳

$$\text{Gain}(S, \text{Wind}) = 0.94 - \frac{6}{14} 1.0 - \frac{8}{14} 0.8113 = 0.0478$$

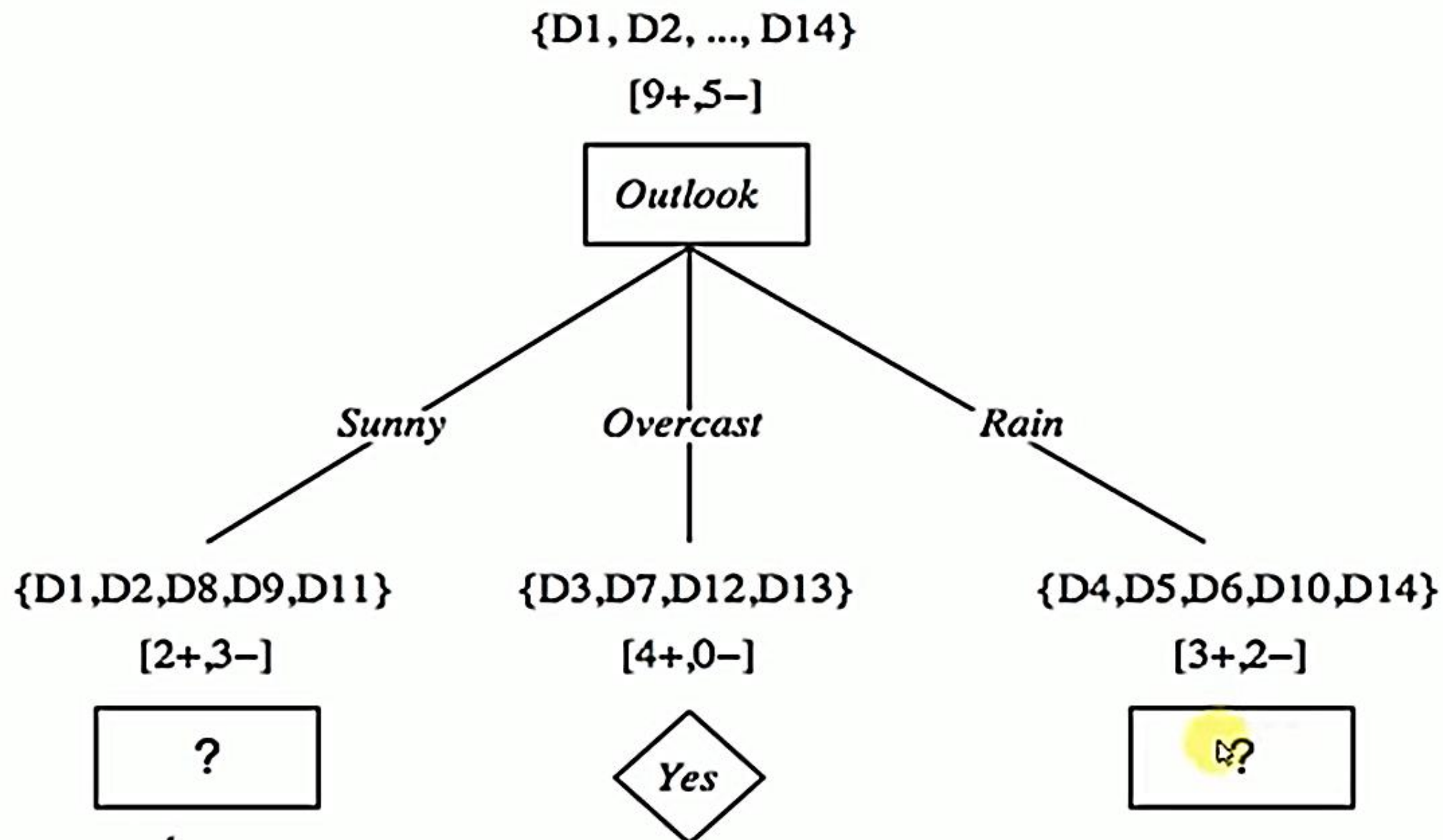
Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$$Gain(S, Outlook) = 0.2464$$

$$Gain(S, Temp) = 0.0289$$

$$Gain(S, Humidity) = 0.1516$$

$$Gain(S, Wind) = 0.0478$$



Day	Temp	Humidity	Wind	Play Tennis
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
D11	Mild	Normal	Strong	Yes

Attribute: Temp

$$\text{Entropy} = -\frac{P}{P+N} \log_2\left(\frac{P}{P+N}\right) - \frac{N}{P+N} \log_2\left(\frac{N}{P+N}\right)$$

Values (Temp) = Hot, Mild, Cool

$$S_{\text{Sunny}} = [2+, 3-]$$

$$\text{Entropy}(S_{\text{Sunny}}) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97$$

$$S_{\text{Hot}} \leftarrow [0+, 2-]$$

$$\text{Entropy}(S_{\text{Hot}}) = 0.0$$

$$S_{\text{Mild}} \leftarrow [1+, 1-]$$

$$\text{Entropy}(S_{\text{Mild}}) = 1.0$$

$$S_{\text{Cool}} \leftarrow [1+, 0-]$$

$$\text{Entropy}(S_{\text{Cool}}) = 0.0$$

$$\text{Gain}(S_{\text{Sunny}}, \text{Temp}) = \text{Entropy}(S) - \sum_{v \in \{\text{Hot}, \text{Mild}, \text{Cool}\}} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

$$\text{Gain}(S_{\text{Sunny}}, \text{Temp})$$

$$= \text{Entropy}(S) - \frac{2}{5} \text{Entropy}(S_{\text{Hot}}) - \frac{2}{5} \text{Entropy}(S_{\text{Mild}})$$

$$- \frac{1}{5} \text{Entropy}(S_{\text{Cool}})$$

$$\text{Gain}(S_{\text{Sunny}}, \text{Temp}) = 0.97 - \frac{2}{5} 0.0 - \frac{2}{5} 1 - \frac{1}{5} 0.0 = 0.570$$

Day	Temp	Humidity	Wind	Play Tennis
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
D11	Mild	Normal	Strong	Yes

Attribute: Humidity

$$\text{Entropy} = -\frac{P}{P+N} \log_2\left(\frac{P}{P+N}\right) - \frac{N}{P+N} \log_2\left(\frac{N}{P+N}\right)$$

Values (Humidity) = High, Normal

$$S_{\text{Sunny}} = [2+, 3-]$$

$$\text{Entropy}(S) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97$$

$$S_{\text{High}} \leftarrow [0+, 3-]$$

$$\text{Entropy}(S_{\text{High}}) = 0.0$$

$$S_{\text{Normal}} \leftarrow [2+, 0-]$$

$$\text{Entropy}(S_{\text{Normal}}) = 0.0$$

$$\text{Gain}(S_{\text{Sunny}}, \text{Humidity}) = \text{Entropy}(S) - \sum_{v \in \{\text{High}, \text{Normal}\}} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

$$\text{Gain}(S_{\text{Sunny}}, \text{Humidity}) = \text{Entropy}(S) - \frac{3}{5} \text{Entropy}(S_{\text{High}}) - \frac{2}{5} \text{Entropy}(S_{\text{Normal}})$$

$$\text{Gain}(S_{\text{Sunny}}, \text{Humidity}) = 0.97 - \frac{3}{5} 0.0 - \frac{2}{5} 0.0 = 0.97$$

Day	Temp	Humidity	Wind	Play Tennis
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
D11	Mild	Normal	Strong	Yes

Attribute: Wind

$$\text{Entropy} = -\frac{P}{P+N} \log_2\left(\frac{P}{P+N}\right) - \frac{N}{P+N} \log_2\left(\frac{N}{P+N}\right)$$

Values (Wind) = Strong, Weak

$$S_{\text{Sunny}} = [2+, 3-]$$

$$\text{Entropy}(S) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97$$

$$S_{\text{Strong}} \leftarrow [1+, 1-]$$

$$\text{Entropy}(S_{\text{Strong}}) = 1.0$$

$$S_{\text{Weak}} \leftarrow [1+, 2-]$$

$$\text{Entropy}(S_{\text{Weak}}) = -\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} = 0.9183$$

$$\text{Gain}(S_{\text{Sunny}}, \text{Wind}) = \text{Entropy}(S) - \sum_{v \in \{\text{Strong}, \text{Weak}\}} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

$$\text{Gain}(S_{\text{Sunny}}, \text{Wind}) = \text{Entropy}(S) - \frac{2}{5} \text{Entropy}(S_{\text{Strong}}) - \frac{3}{5} \text{Entropy}(S_{\text{Weak}})$$

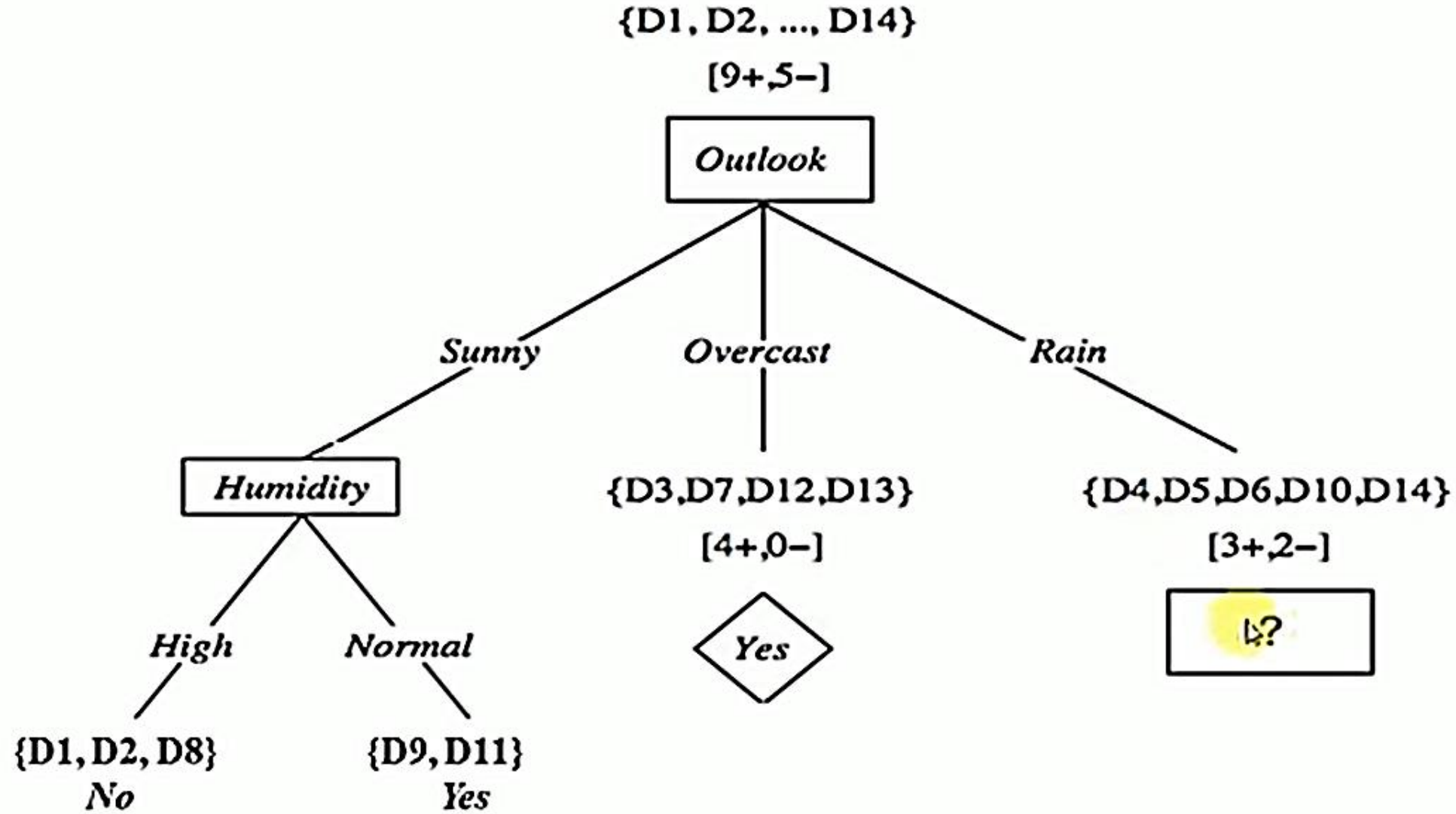
$$\text{Gain}(S_{\text{Sunny}}, \text{Wind}) = 0.97 - \frac{2}{5} 1.0 - \frac{3}{5} 0.918 = 0.0192$$

Day	Temp	Humidity	Wind	Play Tennis
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
D11	Mild	Normal	Strong	Yes

$$Gain(S_{\text{sunny}}, Temp) = 0.570$$

$$Gain(S_{\text{sunny}}, Humidity) = 0.97$$

$$Gain(S_{\text{sunny}}, Wind) = 0.0192$$



Day	Temp	Humidity	Wind	Play Tennis
D4	Mild	High	Weak	Yes
D5	Cool	Normal	Weak	Yes
D6	Cool	Normal	Strong	No
D10	Mild	Normal	Weak	Yes
D14	Mild	High	Strong	No

Attribute: Temp

$$\text{Entropy} = -\frac{P}{P+N} \log_2\left(\frac{P}{P+N}\right) - \frac{N}{P+N} \log_2\left(\frac{N}{P+N}\right)$$

Values (Temp) = Hot, Mild, Cool

$$S_{\text{Rain}} = [3+, 2-]$$

$$\text{Entropy}(S_{\text{Sunny}}) = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.97$$

$$S_{\text{Hot}} \leftarrow [0+, 0-]$$

$$\text{Entropy}(S_{\text{Hot}}) = 0.0$$

$$S_{\text{Mild}} \leftarrow [2+, 1-]$$

$$\text{Entropy}(S_{\text{Mild}}) = -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3} = 0.9183$$

$$S_{\text{Cool}} \leftarrow [1+, 1-]$$

$$\text{Entropy}(S_{\text{Cool}}) = 1.0$$

$$\text{Gain}(S_{\text{Rain}}, \text{Temp}) = \text{Entropy}(S) - \sum_{v \in \{\text{Hot}, \text{Mild}, \text{Cool}\}} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

$$\text{Gain}(S_{\text{Rain}}, \text{Temp})$$

$$= \text{Entropy}(S) - \frac{0}{5} \text{Entropy}(S_{\text{Hot}}) - \frac{3}{5} \text{Entropy}(S_{\text{Mild}})$$

$$- \frac{2}{5} \text{Entropy}(S_{\text{Cool}})$$

$$\text{Gain}(S_{\text{Rain}}, \text{Temp}) = 0.97 - \frac{0}{5} 0.0 - \frac{3}{5} 0.918 - \frac{2}{5} 1.0 = 0.0192$$

Day	Temp	Humidity	Wind	Play Tennis
D4	Mild	High	Weak	Yes
D5	Cool	Normal	Weak	Yes
D6	Cool	Normal	Strong	No
D10	Mild	Normal	Weak	Yes
D14	Mild	High	Strong	No

Attribute: Humidity

$$\text{Entropy} = -\frac{P}{P+N} \log_2\left(\frac{P}{P+N}\right) - \frac{N}{P+N} \log_2\left(\frac{N}{P+N}\right)$$

Values (Humidity) = High, Normal

$$S_{\text{Rain}} = [3+, 2-]$$

$$\text{Entropy}(S_{\text{Sunny}}) = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.97$$

$$S_{\text{High}} \leftarrow [1+, 1-]$$

$$\text{Entropy}(S_{\text{High}}) = 1.0$$

$$S_{\text{Normal}} \leftarrow [2+, 1-]$$

$$\text{Entropy}(S_{\text{Normal}}) = -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3} = 0.9183$$

Q

$$\text{Gain}(S_{\text{Rain}}, \text{Humidity}) = \text{Entropy}(S) - \sum_{v \in \{\text{High}, \text{Normal}\}} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

$$\text{Gain}(S_{\text{Rain}}, \text{Humidity}) = \text{Entropy}(S) - \frac{2}{5} \text{Entropy}(S_{\text{High}}) - \frac{3}{5} \text{Entropy}(S_{\text{Normal}})$$

$$\text{Gain}(S_{\text{Rain}}, \text{Humidity}) = 0.97 - \frac{2}{5} \cdot 1.0 - \frac{3}{5} \cdot 0.918 = 0.0192$$

Day	Temp	Humidity	Wind	Play Tennis
D4	Mild	High	Weak	Yes
D5	Cool	Normal	Weak	Yes
D6	Cool	Normal	Strong	No
D10	Mild	Normal	Weak	Yes
D14	Mild	High	Strong	No

Attribute: Wind

$$\text{Entropy} = -\frac{P}{P+N} \log_2\left(\frac{P}{P+N}\right) - \frac{N}{P+N} \log_2\left(\frac{N}{P+N}\right)$$

Values (wind) = Strong, Weak

$$S_{\text{Rain}} = [3+, 2-]$$

$$\text{Entropy}(S_{\text{Sunny}}) = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.97$$

$$S_{\text{Strong}} \leftarrow [0+, 2-]$$

$$\text{Entropy}(S_{\text{Strong}}) = 0.0$$

$$S_{\text{Weak}} \leftarrow [3+, 0-]$$

$$\text{Entropy}(S_{\text{Weak}}) = 0.0$$

$$\text{Gain}(S_{\text{Rain}}, \text{Wind}) = \text{Entropy}(S) - \sum_{v \in \{\text{Strong}, \text{Weak}\}} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

$$\text{Gain}(S_{\text{Rain}}, \text{Wind}) = \text{Entropy}(S) - \frac{2}{5} \text{Entropy}(S_{\text{Strong}}) - \frac{3}{5} \text{Entropy}(S_{\text{Weak}})$$

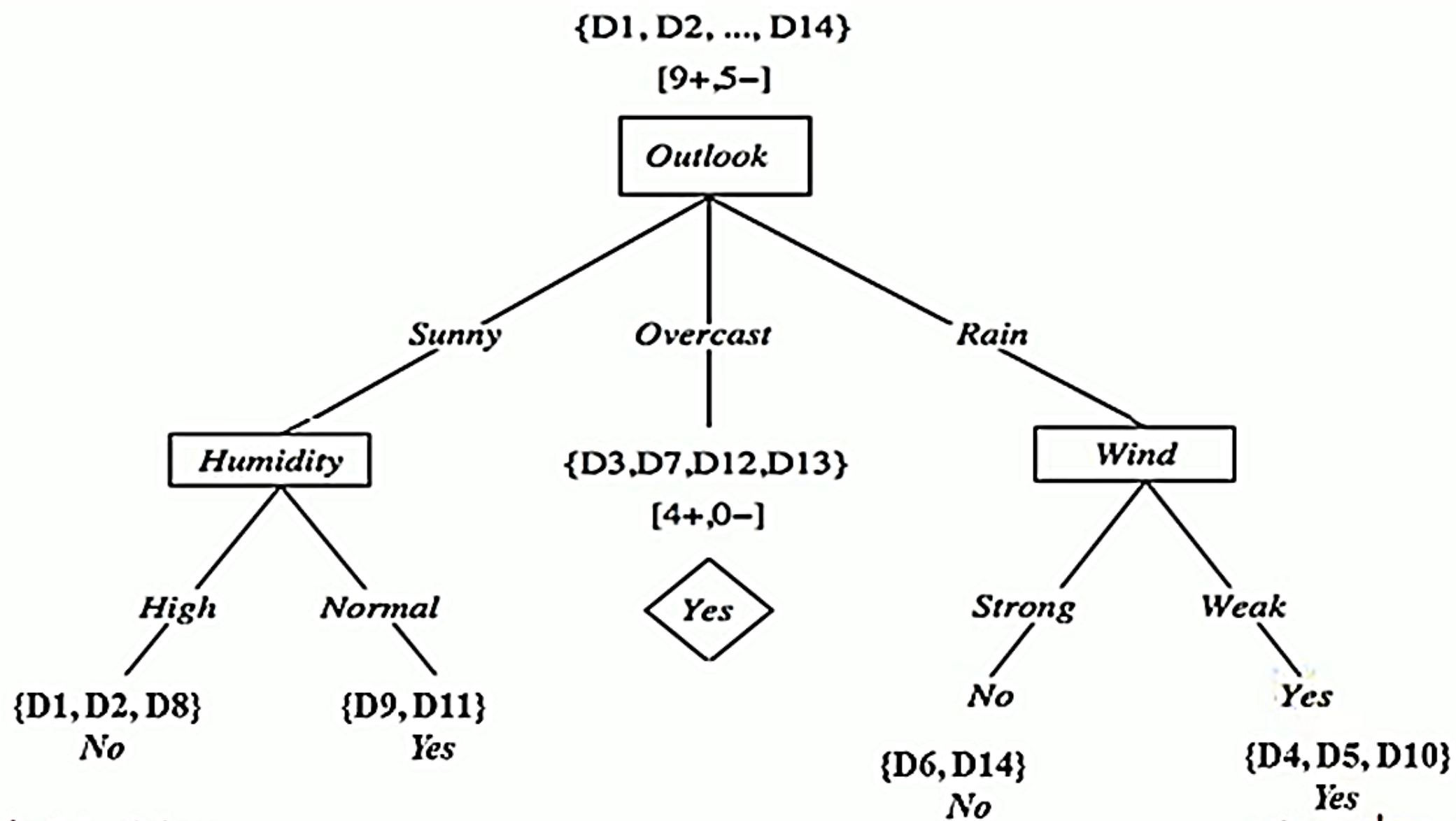
$$\text{Gain}(S_{\text{Rain}}, \text{Wind}) = 0.97 - \frac{2}{5} 0.0 - \frac{3}{5} 0.0 = 0.97$$

Day	Temp	Humidity	Wind	Play Tennis
D4	Mild	High	Weak	Yes
D5	Cool	Normal	Weak	Yes
D6	Cool	Normal	Strong	No
D10	Mild	Normal	Weak	Yes
D14	Mild	High	Strong	No

$$Gain(S_{Rain}, Temp) = 0.0192$$

$$Gain(S_{Rain}, Humidity) = 0.0192$$

$$Gain(S_{Rain}, Wind) = 0.97$$



What is good about Decision Trees

- Interpretable: humans can understand decisions
- Easily handles irrelevant attributes ($G=0$)
- Very compact: # nodes $\ll D$ after pruning
- Very fast at testing time: $O(\text{Depth})$

Limitations for Decision Trees

- Greedy (may not find best tree).
- Instances are represented by attribute/value pairs(e., Outlook: sunny, Wind: strong), but what if we have discrete input values.
- The target function has discrete output values (e.g., Yes, No), thus we cannot have continues number output values.
- The training data may contain errors, or missing attributes
- Uncertainty in the data (e.g., suppose we have two exact days/features, one with “yes” and one with “no”. → no classifier can help in such totally Uncertain data.